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PROFESSIONAL SUMMARY

Interdisciplinary R&D Scientist with a PhD in Systems Biology and extensive experience spanning bioinformatics, molecular biology, and multi-omics analysis. Proven track record in developing automated bioinformatics pipelines and integrating Jupyter-based workflows to accelerate the Design-Build-Test-Learn (DBTL) cycle. Expert in engineering microbial chassis (*P. putida*, *C. glutamicum*) for sustainable bioproduction, combining Genome-scale metabolic modeling with high-throughput experimental validation. Adept at using GitHub for collaborative development in complex, multi-omics environments.

TECHNICAL COMPETENCIES

- **Metabolic Modeling:** COBRA Toolbox, MATLAB, COBRApy, Genome-scale metabolic models, Flux Balance Analysis, Minimal Cut-sets and Elementary Mode Analysis
- **Data Analytics:** Jupyter Notebooks, Excel, Big data analysis, AI Tools - Codex
- **Metabolic Engineering:** Computational strain optimization methods, growth-production tradeoff evaluation
- **Bioprocess Physiology:** Strain physiology in industrially relevant cultivation regimes

PROFESSIONAL EXPERIENCE

Lawrence Berkeley National Laboratory (LBNL)

Research Scientist, Mukhopadhyay Group | August 2021 – July 2026

- **Bioinformatics Pipeline Development:** Implemented automated bioinformatics pipelines utilizing Jupyter Notebooks to process and analyze large-scale multi-omics datasets, reducing data turnaround time and increasing reproducibility.
- **Collaborative Development:** Leveraged GitHub for collaborative projects, ensuring a standardized codebase for analyzing microbial strain physiology, production metrics and omics data-informed genome-scale modeling.
- **Integrated Omics Analysis:** Combined -omics data with physiological readouts to evaluate microbial strain performance under industrially relevant conditions, guiding the design of next-generation engineered strains.
- **Model-Driven Engineering:** Translated complex hypotheses from genome-scale metabolic models into experimental molecular biology designs, resulting in the construction of highly optimized microbial strains for isoprenol and indigoidine production.

- **Data Automation:** Integrated Jupyter-based analysis tools into automated workflows to streamline the transition from raw sequencing/metabolomics data to actionable biological insights.

Joint Bioenergy Institute (JBEI) – Berkeley Lab | Emeryville, California, US

Postdoctoral Fellow, Mukhopadhyay Group | Oct 2018 – Aug 2021

- **Systems Biology Framework:** Developed a scalable, model-integrated workflow to predict and validate strain engineering strategies for biofuels and bioproducts in *Pseudomonas putida* KT2440.
- **Predictive Modeling:** Successfully predicted growth-coupled production strategies for indigoidine; validated these predictions through extensive molecular biology and bioreactor cultivation.
- **NGS Application:** Used genomic characterization tools to verify the stability and efficiency of engineered pathways at scale.

EDUCATION

PhD in Systems Biology | CSIR-National Chemical Laboratory (NCL), Pune, India | 2018

- **Research Focus:** Developed a systems-level understanding of cell metabolism and enzymopathies.
- **Key Project:** Applied constraint-based modeling and molecular biology to study *Chromobacterium violaceum*, identifying key metabolic bottlenecks and proposing NAD⁺/NADH balance strategies to overcome antibiotic resistance.
- **Core Expertise:** Integrated computational predictions with wet-lab validation to map metabolic flux and identify novel targets for strain optimization.

INTELLECTUAL PROPERTY & SELECTED PUBLICATIONS

- **US Patent [11,767,521](#)**- Genetically modified bacterial cells and methods useful for producing indigoidine.
- **US Patent Application 18/976,085:** Genetically modified *Pseudomonas* host cells and methods useful for producing isoprenol. (Co-inventor)
- **Nature Communications (2020):** Lead author on genome-scale metabolic rewiring for indigoidine production at scale. [Link](#)
- **Metabolic Engineering (2024):** Lead author on genome-scale and pathway engineering for sustainable aviation fuel precursors. [Link](#)
- **npj Systems Biology and Applications (2025):** Analysis of genome-scale design tradeoffs in *P. putida*. [Link](#)